

Topic 9: Metals - Answers

IGCSE Chemistry 0620 Paper 4 topical answer booklet.

Subtopic: Properties, extraction, reactivity and corrosion of metals

Questions in this answer PDF: 23

Use this with the matching topical question PDFs. The answer points are organised by the same source labels.

Short explanation notes are added for harder calculation, electrolysis, equilibrium, rate, salts, organic, metals and practical questions.

Periodic table reference pages are not included to save printing paper.

2020 Oct/Nov Paper 41 Question 1

Main topic: Topic 9: Metals | Marks: 9 | Answer source: 0620_w20_ms_41.pdf

Extra topic(s): Topic 2: Atoms, elements and compounds

Answer points

1(a)(i)

- hydrogen / carbon

1(a)(ii)

- silicon

1(a)(iii)

- aluminium

1(a)(iv)

- iron

1(a)(v)

- aluminium

1(a)(vi)

- oxygen

1(b)(i)

- metal higher in reactivity series / metal more reactive (than iron) / allow named metal e.g. magnesium or zinc (1)
- zinc corrodes/oxidises/reacts in preference to iron (1)

1(b)(ii)

- any barrier method e.g. painting

Easy explanation / reminder

- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*
- *Metals tip: link the method to the reactivity series. Very reactive metals need electrolysis; less reactive metal oxides can often be reduced by carbon/CO.*

2020 Oct/Nov Paper 41 Question 2

Main topic: Topic 9: Metals | Marks: 9 | Answer source: 0620_w20_ms_41.pdf

Answer points

2(a)

- zinc blende

2(b)(i)

- $\text{ZnO} + \text{C} \rightarrow \text{Zn} + \text{CO}$
- or
- $2\text{ZnO} + \text{C} \rightarrow 2\text{Zn} + \text{CO}_2$

2(b)(ii)

- chemical change: reduction (1)
- explanation: oxygen is lost (1)

2(b)(iii)

- aluminium is more reactive than carbon

2(b)(iv)

- electrolysis

2(c)

- exists as layers (1)
- (alloy) contains different sized (copper) atoms (1)
- makes it more difficult for layers (of atoms) to slide over each slip/shift other (1)

Easy explanation / reminder

- *Electrolysis tip: cations go to the cathode and gain electrons. Anions go to the anode and lose electrons.*
- *Metals tip: link the method to the reactivity series. Very reactive metals need electrolysis; less reactive metal oxides can often be reduced by carbon/CO.*

2020 Oct/Nov Paper 42 Question 3

Main topic: Topic 9: Metals | Marks: 12 | Answer source: 0620_w20_ms_42.pdf

Extra topic(s): Topic 8: The Periodic Table

Answer points

3(a)

- any two from:
- -
- shiny / lustrous
- -
- conduct electricity
- -
- conduct heat

3(b)

- low(er) density (1)
- low(er) melting points (1)

3(c)(i)

- hydrogen

3(c)(ii)

- hydroxide / OH-

3(c)(iii)

- $2\text{Na} + 2\text{H}_2\text{O} \rightarrow 2\text{NaOH} + \text{H}_2$
- NaOH (as a product) (1)
- rest of equation (1)

3(d)(i)

- galvanising

3(d)(ii)

- prevents water or / and oxygen reaching iron

3(d)(iii)

- zinc more reactive (than iron) (1)
- zinc corrodes / oxidises / reacts in preference to iron (1)

Easy explanation / reminder

- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*
- *Metals tip: link the method to the reactivity series. Very reactive metals need electrolysis; less reactive metal oxides can often be reduced by carbon/CO.*

2020 Oct/Nov Paper 43 Question 1

Main topic: Topic 9: Metals | Marks: 6 | Answer source: 0620_w20_ms_43.pdf

Extra topic(s): Topic 11: Organic chemistry, Topic 7: Acids, bases and salts, Topic 10: Chemistry of the environment, Topic 4: Electrochemistry

Answer points

1(a)

- iron(III) oxide

1(b)

- nitrogen dioxide

1(c)

- ammonia

1(d)

- silver

1(e)

- carbon monoxide

1(f)

- anhydrous cobalt(II) chloride

Easy explanation / reminder

- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*

- Organic tip: identify the functional group first. Combustion of hydrocarbons makes CO₂ and H₂O when oxygen is in excess.

2020 Oct/Nov Paper 43 Question 4

Main topic: Topic 9: Metals | Marks: 17 | Answer source: 0620_w20_ms_43.pdf

Extra topic(s): Topic 3: Stoichiometry

Answer points

4(a)

- sulfur dioxide

4(b)(i)

- Any two from
- (coke)
- -
- releases heat or releases energy(when it reacts with oxygen or burns in air) / (acts as a) fuel / increases temperature (in the furnace) / heats (the furnace) / source of energy
- (coke)
- -
- reduces zinc oxide / is a reducing agent / converts zinc oxide to zinc / removes oxygen from zinc oxide
- (coke)
- -
- (reacts with oxygen) to produce carbon monoxide / reacts with carbon dioxide to form carbon monoxide
- -
- carbon monoxide reduces zinc oxide / converts zinc oxide to zinc / removes oxygen from zinc oxide

4(b)(ii)

- $\text{ZnO} + \text{CO} \rightarrow \text{Zn} + \text{CO}_2$
- OR
- $2\text{ZnO} + \text{C} \rightarrow 2\text{Zn} + \text{CO}_2$
- OR
- $\text{ZnO} + \text{C} \rightarrow \text{Zn} + \text{CO}$

4(b)(iii)

- temperature inside furnace is above / higher than 907 (°C)
- OR temperature is above / higher than the boiling point (of zinc) opposite reasoning also allowed
- OR 1200 (°C) is above / higher than the boiling point (of zinc) opposite reasoning also allowed
- OR 1200 (°C) is above / higher than 907 (°C) opposite reasoning also allowed

4(b)(iv)

- condensation / condensing

4(c)(i)

- zinc

4(c)(ii)

- (a solution containing the) maximum amount of solute dissolved / no more solute can dissolve (1)
- at a given temperature (1)

4(c)(iii)

- one mark for each of any two from:
- -
- zinc oxide
- -
- zinc hydroxide
- -
- zinc carbonate

4(d)(i)

- heat again and weigh again / repeat steps 2 and 3 (1)
- until mass is constant (1)

4(d)(ii)

- (moles of MgSO_4 =) $0.005 / 5 \times 10^{-3}$ (1)
- mass of water = 0.63 g (1)
- moles of water = $0.63 \div 18 = 0.035 / 3.5 \times 10^{-2}$ (1)
- ($x = 0.035 \div 0.005$) = 7 (1)

Easy explanation / reminder

- Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation before converting to the final answer.

- Electrolysis tip: cations go to the cathode and gain electrons. Anions go to the anode and lose electrons.

2021 May/June Paper 41 Question 6

Main topic: Topic 9: Metals | Marks: 13 | Answer source: 0620_s21_ms_41.pdf

Answer points

6(a)(i)

- hematite

6(a)(ii)

- air

6(a)(iii)

- slag / calcium silicate

6(a)(iv)

- any two from:

- -

- (coke)

- releases heat (when it reacts with oxygen or reacts in air)

- OR (acts as a) fuel

- OR increases temperature (in the furnace) / heats (the furnace)

- OR source of energy

- -

- (coke or carbon monoxide)

- reduces iron oxide

- OR is a reducing agent

- OR converts iron oxide to iron /

- removes oxygen from iron oxide

- -

- (coke)

- reacts with oxygen to form carbon monoxide

- OR reacts with carbon dioxide to form carbon monoxide

6(b)

- S²⁻ or S⁻

6(c)(i)

- any two from:

- -

- (iron forms) coloured compounds

- -

- (iron has) variable oxidation states

- -

- (iron acts as a) catalyst

6(c)(ii)

- any two from:

- -

- (iron is) good conductor of electricity

- -

- (iron) forms a basic oxide

- -

- (iron salts are) soluble (in water)

6(d)(i)

- magnesium is more reactive than iron / steel opposite reasoning also allowed (1)

- iron is not oxidised

- OR

- iron does not lose electrons

- OR

- magnesium loses electrons more easily than or in preference (to iron) opposite reasoning also allowed

- OR

- magnesium is oxidised more easily or reacts with oxygen more easily or corrodes more easily or in preference (to iron)

- opposite reasoning also allowed (1)

6(d)(ii)

- copper is less reactive than iron / copper is lower in the reactivity series than iron opposite reasoning also allowed

Easy explanation / reminder

- Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation before converting to the final answer.
- Rate tip: use collision theory. Faster rate means more frequent successful collisions, or more particles with energy greater than activation energy.

2021 May/June Paper 43 Question 6

Main topic: Topic 9: Metals | Marks: 13 | Answer source: 0620_s21_ms_43.pdf

Answer points

6(a)(i)

- limestone

6(a)(ii)

- (molten) iron

6(a)(iii)

- coke / carbon / carbon monoxide

6(a)(iv)

- (reaction between) coke / carbon and oxygen
- Question Answer

6(a)(v)

- nitrogen (1)
- carbon dioxide (1)

6(b)(i)

- zinc sulfide

6(b)(ii)

- condensation/condensing

6(c)

- brass

6(d)(i)

- any two from (copper has):
- -
- high density
- -
- high melting point
- -
- variable oxidation states
- any two from (copper):
- -
- good conductor of electricity
- -
- forms a basic oxide
- -
- soluble salts
- Question Answer

Easy explanation / reminder

- Electrolysis tip: cations go to the cathode and gain electrons. Anions go to the anode and lose electrons.
- Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.

2021 Oct/Nov Paper 41 Question 1

Main topic: Topic 9: Metals | Marks: 18 | Answer source: 0620_w21_ms_41.pdf

Answer points

1(a)(i)

- hydrogen

1(a)(ii)

- sodium

1(a)(iii)

- iron

1(a)(iv)

- aluminium

1(a)(v)

- zinc or copper

1(a)(vi)

- iron

1(b)(i)

- zinc blende

1(b)(ii)

- bauxite

1(c)

- protective oxide layer

1(d)

- Any 2 from 3

--

- zinc

--

- iron

--

- copper

1(e)(i)

- copper

1(e)(ii)

- zinc sulfate

1(e)(iii)

- particles have more energy
- More collisions (between particles) occur per second / per unit time
- greater percentage / proportion / fraction of collisions (of particles) are successful / have energy above activation
- energy / have energy equal to activation energy

1(e)(iv)

- Any 2 from 3:

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- use a catalyst

--

- use smaller granules

--

- increase concentration

Easy explanation / reminder

- *Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation before converting to the final answer.*
- *Rate tip: use collision theory. Faster rate means more frequent successful collisions, or more particles with energy greater than activation energy.*

2022 May/June Paper 41 Question 1

Main topic: Topic 9: Metals | Marks: 9 | Answer source: 0620_s22_ms_41.pdf

Extra topic(s): Topic 11: Organic chemistry

Answer points**1(a)**

- carbon dioxide

1(b)

- aluminium oxide

1(c)

- ethanol (1)
- carbon dioxide (1)

1(d)

- ethanol

1(e)

- oxygen

1(f)

- carbon dioxide

1(g)

- nitrogen

1(h)

- diamond

Easy explanation / reminder

- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*
- *Metals tip: link the method to the reactivity series. Very reactive metals need electrolysis; less reactive metal oxides can often be reduced by carbon/CO.*

2022 May/June Paper 42 Question 1

Main topic: Topic 9: Metals | Marks: 8 | Answer source: 0620_s22_ms_42.pdf

Extra topic(s): Topic 10: Chemistry of the environment, Topic 8: The Periodic Table

Answer points

1(a)

- Mg

1(b)

- Ar

1(c)

- Cl

1(d)

- Si

1(e)

- P

1(f)

- Na

1(g)

- Al

1(h)

- S

Easy explanation / reminder

- *Metals tip: link the method to the reactivity series. Very reactive metals need electrolysis; less reactive metal oxides can often be reduced by carbon/CO.*

2023 May/June Paper 42 Question 1

Main topic: Topic 9: Metals | Marks: 6 | Answer source: 0620_s23_ms_42.pdf

Extra topic(s): Topic 10: Chemistry of the environment

Answer points

1(a)

- D

1(b)

- C

1(c)

- H

1(d)

- B

1(e)

- C

1(f)

- G

Easy explanation / reminder

- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*
- *Metals tip: link the method to the reactivity series. Very reactive metals need electrolysis; less reactive metal oxides can often be reduced by carbon/CO.*

2023 Oct/Nov Paper 41 Question 3

Main topic: Topic 9: Metals | Marks: 12 | Answer source: 0620_w23_ms_41.pdf

Extra topic(s): Topic 8: The Periodic Table

Answer points

3(a)(i)

- Ag column all ✗ (1)
- ✗ in Pb AND 2 ✓ in Zn (1)
- Zn, Mn, Pb Ag (1)

3(a)(ii)

- (all) nitrates are soluble OR lead sulfate is insoluble

3(a)(iii)

- $\text{Zn} + 2\text{AgNO}_3 \rightarrow \text{Zn}(\text{NO}_3)_2 + 2\text{Ag}$
- $\text{Zn}(\text{NO}_3)_2$ on the right hand side (1)
- correct equation (1)

3(b)(i)

- colourless (1)
- orange (1)

3(b)(ii)

- $\text{Cl}_2(\text{g}) + 2\text{Br}^-(\text{aq}) \rightarrow \text{Br}_2(\text{aq}) + 2\text{Cl}^-(\text{aq})$
- $\text{Br}_2 + \text{Cl}^-$ as products (1)
- correct equation (1)
- state symbols (1)

3(b)(iii)

- tenessine / Ts

2023 Oct/Nov Paper 42 Question 3

Main topic: Topic 9: Metals | Marks: 17 | Answer source: 0620_w23_ms_42.pdf

Answer points

3(a)

- (hot) air

3(b)(i)

- hematite

3(b)(ii)

- $\text{Fe}_2\text{O}_3 + 3\text{CO} \rightarrow 3\text{CO}_2 + 2\text{Fe}$

3(b)(iii)

- from +3 (1)
- to 0 (1)

3(b)(iv)

- decrease (in oxidation number)

3(c)

- $\text{CaCO}_3 \rightarrow \text{CaO} + \text{CO}_2$ (1)
- thermal decomposition (1)
- $\text{SiO}_2 + \text{CaO} \rightarrow \text{CaSiO}_3$ (1)
- acid-base reaction (1)

3(d)(i)

- alloy

3(d)(ii)

- nickel / chromium

3(e)

- hydrated iron(III) oxide

3(f)(i)

- galvanising

3(f)(ii)

- painting / greasing / coating with plastic

3(f)(iii)

- zinc more reactive (than iron) (1)
- zinc corrodes / oxidises / reacts in preference to / instead of iron (1)

Easy explanation / reminder

- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*
- *Metals tip: link the method to the reactivity series. Very reactive metals need electrolysis; less reactive metal oxides can often be reduced by carbon/CO.*

2023 Oct/Nov Paper 43 Question 5

Main topic: Topic 9: Metals | Marks: 18 | Answer source: 0620_w23_ms_43.pdf

Answer points

5(a)(i)

- positive ions / cations (1)
- sea of electrons / mobile electrons / delocalised electrons (1)
- attraction between positive ions and electrons (1)

5(a)(ii)

- electrons move / electrons mobile / electrons flow

5(b)(i)

- any two
- -
- (coke) releases heat
- -
- (coke) reduces iron(III) oxide
- -
- (coke) reacts with carbon dioxide to form carbon monoxide

5(b)(ii)

- limestone decomposes to calcium oxide (1)
- calcium oxide reacts with / neutralises silicon(IV) oxide to produce slag / calcium silicate (1)

5(b)(iii)

- $\text{Fe}_2\text{O}_3 + 3\text{C} \rightarrow 2\text{Fe} + 3\text{CO}$
- OR
- $\text{Fe}_2\text{O}_3 + 3\text{CO} \rightarrow 2\text{Fe} + 3\text{CO}_2$
- OR
- $2\text{Fe}_2\text{O}_3 + 3\text{C} \rightarrow 4\text{Fe} + 3\text{CO}_2$
- all formulae (1)
- equation correct (1)

5(b)(iv)

- temperature in the furnace is above the melting point of iron

5(c)

- atoms have different sizes (1)
- layers cannot slide over or past each other (1)

5(d)(i)

- oxygen AND water

5(d)(ii)

- galvanising

5(d)(iii)

- acts as a barrier which prevents oxygen and water reaching the iron

5(d)(iv)

- zinc is more reactive than iron (1)
- zinc is oxidised / zinc loses electrons / zinc forms positive ions / forms zinc ions (1)

Easy explanation / reminder

- *Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation before converting to the final answer.*
- *Electrolysis tip: cations go to the cathode and gain electrons. Anions go to the anode and lose electrons.*

2024 Feb/March Paper 42 Question 1

Main topic: Topic 9: Metals | Marks: 19 | Answer source: 0620_m24_ms_42.pdf

Answer points

1(a)

- coke

1(a)(ii)

- provide heat

1(b)(i)

- 160
- 112 AND 70.0(0)(%)

1(b)(ii)

- hematite

1(b)(iii)

- by reduction of carbon dioxide

1(b)(iv)

- $\text{Fe}_2\text{O}_3 + 3\text{CO} \rightarrow 2\text{Fe} + 3\text{CO}_2$

- species

- correct equation

1(b)(v)

- reduction

1(c)

- thermal decomposition

1(d)(i)

- CaO is basic

- SiO₂ is acidic

1(d)(ii)

- silicon(IV) oxide

- slag

1(e)(i)

- aluminium is above carbon in the reactivity series

- OR

- aluminium is more reactive than carbon

- February/March 2024

1(e)(ii)

- electrolysis

1(f)(i)

- 2,8

1(f)(ii)

- 26 (protons)

- 23 (electrons)

Easy explanation / reminder

- *Electrolysis tip: cations go to the cathode and gain electrons. Anions go to the anode and lose electrons.*

- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*

2024 May/June Paper 41 Question 6

Main topic: Topic 9: Metals | Marks: 14 | Answer source: 0620_s24_ms_41.pdf

Answer points

6(a)(i)

- limestone

- OR

- calcium carbonate

6(a)(ii)

- (molten) iron

6(a)(iii)

- silicon(IV) oxide

- OR

- silicon dioxide

6(a)(iv)

- coke or carbon and oxygen

6(a)(v)

- Any two from:

- ■

- nitrogen

- ■

- carbon dioxide

- ■

- argon

6(b)(i)

- the temperature in the furnace is above or higher than the boiling point of zinc opposite reasoning also allowed

- OR
- the boiling point of zinc is below or less than the temperature of the furnace opposite reasoning also allowed

6(b)(ii)

- condensation / condensing

6(c)(i)

- galvanising

6(c)(ii)

- zinc is more reactive than iron (1)
- zinc is oxidised /
- zinc loses electrons /
- zinc forms positive ions /
- zinc forms zinc ions (1)

6(d)(i)

- amphoteric

6(d)(ii)

- Na_2ZnO_2

6(d)(iii)

- zinc sulfate

Easy explanation / reminder

- *Electrolysis tip: cations go to the cathode and gain electrons. Anions go to the anode and lose electrons.*
- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*

2024 May/June Paper 42 Question 2

Main topic: Topic 9: Metals | Marks: 14 | Answer source: 0620_s24_ms_42.pdf

Extra topic(s): Topic 7: Acids, bases and salts

Answer points

2(a)(i)

- electron loss (from outer shell)

2(a)(ii)

- potassium

2(a)(iii)

- magnesium or potassium

2(a)(iv)

- iron

2(b)

- Colourless
- Blue
- Colour given in
-

2(c)(i)

- hydrogen

2(c)(ii)

- strontium hydroxide

2(c)(iii)

- magnesium or Mg
- magnesium oxide

2(d)

- $\text{Fe}_3\text{O}_4 + 8\text{HCl} \rightarrow 2\text{FeCl}_3 + \text{FeCl}_2 + 4\text{H}_2\text{O}$
- FeCl_2 or FeCl_3 as product of equation
- FeCl_2 and FeCl_3 and H_2O as only products of equation
- Correct equation

Easy explanation / reminder

- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*

2024 May/June Paper 43 Question 6

Main topic: Topic 9: Metals | Marks: 13 | Answer source: 0620_s24_ms_43.pdf

Answer points

6(a)(i)

- hematite

6(a)(ii)

- air

6(a)(iii)

- carbon monoxide

6(a)(iv)

- limestone thermally decomposes (into calcium oxide and carbon dioxide)
- calcium oxide reacts with silicon(IV) oxide
- producing slag

6(b)(i)

- have low densities
- or
- have low melting point

6(b)(ii)

- act as catalysts
- or
- form coloured compounds

6(b)(iii)

- form basic oxides
- or
- form positive ions

6(b)(iv)

- form acidic oxides
- or
- form negative ions

6(c)(i)

- magnesium loses electrons more readily than / in preference to / instead of iron
- magnesium is more reactive than iron opposite reasoning also allowed

6(c)(ii)

- any metal below iron in the reactivity series e.g. copper / silver / gold

Easy explanation / reminder

- *Rate tip: use collision theory. Faster rate means more frequent successful collisions, or more particles with energy greater than activation energy.*
- *Electrolysis tip: cations go to the cathode and gain electrons. Anions go to the anode and lose electrons.*

2024 Oct/Nov Paper 43 Question 1

Main topic: Topic 9: Metals | Marks: 9 | Answer source: 0620_w24_ms_43.pdf

Extra topic(s): Topic 2: Atoms, elements and compounds, Topic 11: Organic chemistry, Topic 4: Electrochemistry, Topic 7: Acids, bases and salts

Answer points**1(a)**

- ethanol / carbon dioxide

1(b)

- helium

1(c)

- carbon dioxide

1(d)

- cryolite

1(e)

- hematite

1(f)

- hydrogen

1(g)

- silicon(IV) oxide

1(h)

- graphite

1(i)

- carbon dioxide

Easy explanation / reminder

- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*
- *Metals tip: link the method to the reactivity series. Very reactive metals need electrolysis; less reactive metal oxides can often be reduced by carbon/CO.*

2025 May/June Paper 41 Question 3

Main topic: Topic 9: Metals | Marks: 15 | Answer source: 0620_s25_ms_41.pdf

Extra topic(s): Topic 7: Acids, bases and salts

Answer points

3(a)(i)

- positive ions / cations(1)
- sea of electrons /
- mobile electrons /
- delocalised electrons(1)
- attraction between positive ions and electrons(1)

3(a)(ii)

- electrons move /
- electrons mobile /
- electrons flow

3(b)(i)

- harder or stronger

3(b)(ii)

- zinc

3(c)(i)

- Any two from:-
- -
- solid disappears / solid dissolves
- -
- fizzing / bubbling / effervescence
- -
- blue solution

3(c)(ii)

- all the sulfuric acid had reacted

3(c)(iii)

- copper(II) carbonate

3(c)(iv)

- copper(II) oxide / copper(II) hydroxide

3(c)(v)

- saturated (solution)

3(c)(vi)

- more particles per unit volume
- more collisions per unit time / increased collision frequency

3(c)(vii)

- hydrated

Easy explanation / reminder

- *Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation before converting to the final answer.*
- *Rate tip: use collision theory. Faster rate means more frequent successful collisions, or more particles with energy greater than activation energy.*

2025 May/June Paper 42 Question 4

Main topic: Topic 9: Metals | Marks: 9 | Answer source: 0620_s25_ms_42.pdf

Extra topic(s): Topic 3: Stoichiometry

Answer points

4(a)(i)

- 4CO

4(a)(ii)

- chromium will need to be separated from iron

4(b)

- alloy

4(c)(i)

- -2

4(c)(ii)

- use of -14 or 7 → -2
- +6

4(c)(iii)

- mol $(\text{NH}_4)_2\text{Cr}_2\text{O}_7 = 1.26 / 252 = 0.005(00)$
- mol of $\text{N}_2 = 0.005(00)$
- vol of $\text{N}_2 = 0.005 \rightarrow 24000 = 120 \text{ (cm}^3\text{)}$

Easy explanation / reminder

- *Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation before converting to the final answer.*
- *Metals tip: link the method to the reactivity series. Very reactive metals need electrolysis; less reactive metal oxides can often be reduced by carbon/CO.*

2025 May/June Paper 43 Question 3

Main topic: Topic 9: Metals | Marks: 15 | Answer source: 0620_s25_ms_43.pdf

Extra topic(s): Topic 7: Acids, bases and salts, Topic 2: Atoms, elements and compounds

Answer points**3(a)(i)**

- metallic (bonding)

3(a)(ii)

- positive ions
- electrons

3(a)(iii)

- electrons

3(b)(i)

- a mixture of a metal with other elements

3(b)(ii)

- copper

3(c)(i)

- solid stops dissolving / disappearing
- no more bubbles / fizzing / effervescence on addition of solid

3(c)(ii)

- aqueous zinc sulfate

3(c)(iii)

- zinc oxide
- OR
- zinc hydroxide

3(c)(iv)

- (a solution containing the) maximum concentration of a solute dissolved in the solvent
- at a specified temperature

3(c)(v)

- (large pieces have) smaller surface area
- frequency of collisions between (zinc carbonate and acid) particles decreases

3(c)(vi)

- (crystals that are) chemically combined with water

Easy explanation / reminder

- *Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation before converting to the final answer.*
- *Rate tip: use collision theory. Faster rate means more frequent successful collisions, or more particles with energy greater than activation energy.*

2025 Oct/Nov Paper 42 Question 2

Main topic: Topic 9: Metals | Marks: 19 | Answer source: 0620_w25_ms_42.pdf

Extra topic(s): Topic 2: Atoms, elements and compounds, Topic 6: Chemical reactions

Answer points

2(a)(i)

- hematite

2(a)(ii)

- burning of carbon (coke)

2(a)(iii)

- carbon monoxide

2(a)(iv)

- $3\text{CO} + \text{Fe}_2\text{O}_3 \rightarrow 2\text{Fe} + 3\text{CO}_2$
- species
- correct equation

2(a)(v)

- $\text{CaCO}_3 \rightarrow \text{CaO} + \text{CO}_2$
- $\text{CaO} + \text{SiO}_2 \rightarrow \text{CaSiO}_3$

2(b)

- atoms of the same element that have the same number of protons
- but different numbers of neutrons

2(c)(i)

- $p = 26$ and 26
- $n = 28$ and 32
- $e = 24$ and 23

2(c)(ii)

- NaOH(aq) OR $\text{NH}_3\text{(aq)}$
- green ppt.
- (insoluble in excess, ppt. turns brown near surface on standing)

2(d)(i)

- a substance that oxidises another substance and is itself reduced
- 'a substance' oxidises another substance
- is itself reduced

2(d)(ii)

- loss of (one) electron

2(d)(iii)

- acidified

2(e)

- iodine

Easy explanation / reminder

- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*
- *Metals tip: link the method to the reactivity series. Very reactive metals need electrolysis; less reactive metal oxides can often be reduced by carbon/CO.*